

Accurate modeling for 3×2pt analyses in Roman and Rubin: a study of model approximations

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One of the pillars of modern cosmology is the use of galaxy imaging surveys to extract information from the large-scale structure. In recent surveys, this measurement is typically carried out via a combination of auto and cross-correlations between galaxy density and galaxy weak lensing, an analysis referred to as the “3×2pt” analysis. In this paper, we carry out a systematic study of three modeling approximations sometimes used in a 3×2pt analysis: 1) applying the Limber approximation, 2) neglecting redshift-space distortion (RSD), and 3) using less accurate models for the nonlinear matter power spectrum. We study the problem in the context of the final data from two major Stage-IV galaxy imaging surveys: the Nancy Grace Roman Space Telescope’s High Latitude Imaging Survey and the Vera C. Rubin Observatory’s Legacy Survey of Space and Time. To do this, we first validate our modeling pipeline in COCOA with an established code base CCL. Next we perform a simulated likelihood analysis to assess the impact of these approximations. We find all three effects to be important; neglecting any of them can induce biases in cosmological constraints greater than 1σ . Moreover, we explore how the lens galaxy sample configuration and scale cut can influence the constraint and inference.

I. INTRODUCTION

Extracting large-scale structure (LSS) information from galaxy imaging surveys is one of the main leverages we have to test the current Λ CDM cosmology paradigm. Over the past 10 years, the community has established a fairly mature framework to achieve this goal – using a combination of auto- and cross-correlations of galaxy density and galaxy weak lensing. Concretely, we use three two-point correlation functions: the auto-correlation of galaxy weak lensing (cosmic shear), the auto-correlation of galaxy density (galaxy clustering), and the cross-correlation of galaxy position and galaxy shear (galaxy-galaxy lensing). This combination, commonly referred to as “3×2pt” as it is a combination of three two-point functions, has been carried out by all Stage-III galaxy imaging surveys [9, 17, 40].

There are several advantages in combining these three two-point functions. First, it increases the overall statistical power by breaking degeneracies in the cosmological and nuisance parameters. Second, there are shared measurement systematic nuisance parameters between the three two-point functions, allowing the data to self-calibrate these quantities. Finally, requiring the three two-point functions to be consistent with each other places a very strong test on the data – if there is residual systematic effects in one of the probes, it could show up as being incompatible with the rest of the probes and trigger further investigation. The recent Dark Energy Survey Year 6 (DES-Y6) 3×2pt analysis [9] nicely demonstrates this point in that the 3×2pt constraints appear to be much more robust to modeling choices and other assumptions of systematic effects.

In fact, the development of the 3×2pt framework is likely one of the main lessons and legacy from the DES.

With the wrapping up of the Stage-III surveys, we are now facing new challenges from the next generation galaxy imaging surveys such as the Nancy Grace Roman Space Telescope High Latitude Imaging Survey¹ (HLIS), the Rubin Observatory’s Legacy Survey of Space and Time² (LSST) and the Euclid mission³. These surveys are expected to provide unprecedented statistical power, which places unprecedented stringent requirements on our tolerance of systematic effects. As such, it is important to examine all aspects of the 3×2pt analysis and ensure that the pipeline we used for Stage-III surveys remain valid.

In this work, we focus on three specific modeling components related to the galaxy and matter power spectrum underlying the 3×2pt analysis. First, we look at the impact of introducing the Limber approximation to the probes. Second, we examine the impact of including redshift-space distortion (RSD) to the probes. Finally, we look at how the 3×2pt results are sensitive to the choice of model for the nonlinear matter power spectrum. We choose to study these three effects as these are effects where there is a known “truth” or more accurate modeling option, unlike effects such as intrinsic alignment, which at the moment we do not accurately model.

¹ <https://roman-hlis-cosmology.caltech.edu/>

² <https://rubinobservatory.org/>

³ https://www.esa.int/Science_Exploration/Space_Science/Euclid

Several studies have investigated these approximation in the past. For Limber approximation, [21] showed that the usual first-order Limber approximation is insufficiently accurate for the full 10-year (Y10) LSST dataset on $\ell = 200 - 1000$. [13] showed that using the non-Limber model can effectively correct the biases induced by the Limber approximation. They also showed that a full non-Limber model of galaxy clustering is necessary for the first year (Y1) of LSST 3 $\times$ 2pt analysis not to be significantly biased. For the nonlinear power spectrum, [28] showed that using HMCode instead of the EUCLIDEMULATOR2 results in negligible shift in cosmology when applied to the DES-Y6 3 $\times$ 2pt analysis. In [34], the authors suggest that the same conclusion holds for cosmic shear-only in a Euclid-like survey. For RSD, [35] showed that when the linear RSD is neglected, significant biases occur in Λ CDM with the final 6-year Euclid dataset.

The main goal of this paper is to perform an updated and systematic study of these three effects in the context of a 3 $\times$ 2pt analysis for the final 5-year (Y5) Roman and 10-year (Y10) Rubin dataset. By doing this, we also wish to achieve two secondary goals: First, we will update some of the analysis choices that incorporates our knowledge from Stage-III surveys. Second, we carry out a code comparison between the official modeling packages in Roman (CocOA) and Rubin (CCL), ensuring the robustness of the modeling framework across the Stage-IV surveys.

The paper is structured as follows. In Section II A we briefly describe our modeling framework for the 3 $\times$ 2pt data vector and highlight the three approximations we will focus on. In Section III we describe the data configurations we choose to use for both Roman and Rubin's 3 $\times$ 2pt analysis. In Section IV, we introduce our simulated likelihood analysis. In Section V we present our results and recommendation for the two surveys. We finally conclude in Section VI.

II. MODELING AND APPROXIMATIONS

In this section we first describe the overall framework of modeling the 3 $\times$ 2pt data vector in Section II A and then zoom into describing the three approximations we will be studying in this work in Section II B.

In all our tests, we consider a Λ CDM model, which assumes dark energy is a cosmological constant and has 5 free parameters ($A_s, n_s, H_0, \Omega_b, \Omega_m$). Here, A_s and n_s are the amplitude and scalar spectral index of the primordial curvature perturbation power spectrum, H_0 is the Hubble constant, and Ω_m and Ω_b are the matter density parameter and the baryon density parameter.

A. Modeling 3 $\times$ 2pt data vector

1. Power spectrum

We follow the formalism used in the DES-Y6 analysis [28] in modeling the 3 $\times$ 2pt data vector. We focus on two projected fields: the projected galaxy overdensity field (δ_g) probed by

the lens galaxy sample and the weak lensing shear field ($\gamma = \gamma_1 + i\gamma_2$) probed by the source galaxy sample. For a given observation direction $\hat{n}$, they are described below without any correction term,

$$\delta_{g,\text{obs}}^i = \delta_{g,D}^i(\hat{n}) = \int d\chi W_{\delta,g}^i \delta_g^{(3D)}(\hat{n}\chi, \chi), \quad (1)$$

$$\gamma_1^i = \gamma_{1,G}^i(\hat{n}) = \int d\chi W_{\kappa,g}^i \chi (\Phi_{,11} - \Phi_{,22})(\hat{n}\chi, \chi), \quad (2)$$

$$\gamma_2^i = \gamma_{2,G}^i(\hat{n}) = \int d\chi W_{\kappa,g}^i \chi (\Phi_{,12} + \Phi_{,21})(\hat{n}\chi, \chi). \quad (3)$$

The subscript (g) represents the galaxy samples which could be either lens samples (l) or source samples (s) depending on the observable calculated. They are divided into different tomographic bins and the superscript (i) denotes the bin indices. $\delta_{g,D}$ denotes the line-of-sight projection component of $\delta_g^{(3D)}$, the three-dimensional galaxy density contrast field. $\gamma_{\alpha,G}$ ($\alpha = 1, 2$) denotes the gravitational shear component, directly induced by the $\Phi_{,\alpha}$, the spatial transverse derivative of the 3D potential Φ . $W_{\delta,g}^i$ and $W_{\kappa,g}^i$ are the normalized selection function and the lensing efficiency, respectively. They are defined as,

$$W_{\delta,g}^i = n_g^i(z(\chi)) \frac{dz}{d\chi}, \quad (4)$$

$$W_{\kappa,g}^i = \frac{3\Omega_m H_0^2}{2c^2} \int_{\chi}^{\infty} d\chi' n_g^i(\chi') \frac{\chi}{a(\chi)} \frac{\chi' - \chi}{\chi'}, \quad (5)$$

where $n_g^i(z)$ is the normalized redshift distribution of the galaxy sample in tomographic bin i .

We can then express the angular power spectrum of two projected fields by computing the second moment of their spherical harmonic coefficients. For example, the angular power spectrum of two projected galaxy overdensity fields is,

$$\begin{aligned} C_{\delta_g,D \delta_g,D}^{ij} &= \langle a_{\ell m}^i a_{\ell' m'}^{j*} \rangle / (\delta_{D,\ell\ell'} \delta_{D,mm'}) \\ &= (4\pi)^2 \int d\chi_1 W_{\delta,g}^i(\chi_1) \int d\chi_2 W_{\delta,g}^j(\chi_2) \\ &\quad \cdot \int \frac{dk_1^3}{(2\pi)^3} j_{\ell}(k_1 \chi_1) Y_{\ell m}(\hat{\mathbf{k}}_1) \\ &\quad \cdot \int \frac{dk_2^3}{(2\pi)^3} j_{\ell}(k_2 \chi_2) Y_{\ell m}^*(\hat{\mathbf{k}}_2) \langle \tilde{\delta}_g^{(3D)}(\mathbf{k}_1, \chi_1), \tilde{\delta}_g^{(3D)}(\mathbf{k}_2, \chi_2) \rangle \\ &= \frac{2}{\pi} \int d\chi_1 W_{\delta,g}^i(\chi_1) \int d\chi_2 W_{\delta,g}^j(\chi_2) \\ &\quad \cdot \int k^2 dk j_{\ell}(k \chi_1) j_{\ell}(k \chi_2) P_{gg}(\mathbf{k}, \chi_1, \chi_2). \end{aligned} \quad (6)$$

where the relation,

$$\langle \tilde{\delta}_g^{(3D)}(\mathbf{k}_1, \chi_1), \tilde{\delta}_g^{(3D)}(\mathbf{k}_2, \chi_2) \rangle = (2\pi)^3 \delta_D(\mathbf{k}_1 - \mathbf{k}_2) P_{gg}(\mathbf{k}, \chi_1, \chi_2), \quad (7)$$

is used and δ_D is the Dirac delta function, j_{ℓ} is the spherical Bessel function, and P_{gg} is the unequal time galaxy power spectrum. The superscript i and j denote the indices of the

tomographic bins, while $\chi_1, \chi_2, \mathbf{k}_1, \mathbf{k}_2$ are dummy variables used to distinguish the two line-of-sight integrations. The angular power spectrum of cosmic shear and galaxy-galaxy lensing can be derived following the same math.

2. correlation function

Correlation functions in the real space can be transformed from the angular power spectrum. We describe this relation following [28],

$$\begin{aligned} w^{ij}(\theta) &= \sum_{\ell} \frac{2\ell+1}{4\pi} P_{\ell}(\cos \theta) C_{\delta_{g,\text{obs}}, \delta_{g,\text{obs}}}^{ij}(\ell), \\ \gamma_t^{ij}(\theta) &= \sum_{\ell} \frac{2\ell+1}{4\pi} \frac{P_{\ell}^2(\cos \theta)}{\ell(\ell+1)} C_{\delta_{g,\text{obs}}, E}^{ij}(\ell), \\ \xi_{\pm}^{ij}(\theta) &= \sum_{\ell} \frac{2\ell+1}{4\pi} \frac{2(G_{\ell,2}^{+}(x) \pm G_{\ell,2}^{-}(x))}{\ell^2(\ell+1)^2} [C_{EE}^{ij}(\ell) \pm C_{BB}^{ij}(\ell)], \end{aligned} \quad (8)$$

where P_{ℓ} is the Legendre polynomials and the G_{ℓ} is the associate Legendre polynomials. E and B refers to the E- and B-modes of the shear field, defined as

$$\gamma_{E/B}^i(\ell) = (\delta_{\alpha\beta}, \epsilon_{\alpha\beta}) T_{\alpha}(\ell) \gamma_{\beta}^i(\ell), \quad (9)$$

where $\mathbf{T}(\ell) \equiv (\cos(2\phi_{\ell}), \sin(2\phi_{\ell}))$, ϕ_{ℓ} is the angle of the ℓ vector from the ℓ_x axis, $\delta_{\alpha\beta}$ is the Dirac-delta function, and $\epsilon_{\alpha\beta}$ is the two-dimensional Levi-Civita tensor.

It is impossible to obtain the correlation function at a precise angular point because of finite galaxy samples, so it is measured over an angular bin $[\theta_{\min}, \theta_{\max}]$. To match what we do in real galaxy surveys, correlation functions need to be averaged over angular bins, i.e. bin-average correction. It can be done by replacing these Legendre functions with their bin-averaged versions $\overline{P}_{\ell}$, $\overline{P}_{\ell}^2$, and $\overline{G_{\ell,2}^{+} \pm G_{\ell,2}^{-}}$. The example of the bin-averaged versions for P_{ℓ} is,

$$\begin{aligned} \overline{P}_{\ell}(\theta_{\min}, \theta_{\max}) &\equiv \frac{\int_{\cos \theta_{\min}}^{\cos \theta_{\max}} dx P_{\ell}(x)}{\cos \theta_{\max} - \cos \theta_{\min}} \\ &= \frac{[P_{\ell+1}(x) - P_{\ell-1}(x)]|_{\cos \theta_{\min}}^{\cos \theta_{\max}}}{(2\ell+1)(\cos \theta_{\max} - \cos \theta_{\min})}. \end{aligned} \quad (10)$$

For small sky patches, it is computationally efficient to replace the brute-force summation of the Legendre function by the integration of the first kind Bessel function without losing too much precision, i.e. the flat sky approximation.

$$w^{ij}(\theta) = \int_0^{\infty} \frac{d\ell}{2\pi} C_{\delta_{g,\text{obs}}, \delta_{g,\text{obs}}}^{ij} J_0(\ell\theta), \quad (11)$$

$$\gamma_t^{ij}(\theta) = \int_0^{\infty} \frac{d\ell}{2\pi} C_{\delta_{g,\text{obs}}, E}^{ij} J_2(\ell\theta), \quad (12)$$

$$\xi_{\pm}^{ij}(\theta) = \int_0^{\infty} \frac{d\ell}{2\pi} C_{EE}^{ij} J_{0/4}(\ell\theta). \quad (13)$$

On top of the basic modeling of the tracers, we also include the following components

• Intrinsic alignment (IA)

The gravitational shear field $\gamma_{\alpha,G}$ is correlated with the galaxy intrinsic shear field or the intrinsic ellipticity $\gamma_{\alpha,I}$. Thus, the latter has a non-negligible contribution to the cosmic shear signal and should be taken into account in the weak lensing shear field in Eqs. 2 and 3.

$$\gamma_{\alpha} = \gamma_{\alpha,G} + \gamma_{\alpha,I}. \quad (14)$$

We parameterize and marginalize out this effect using the Nonlinear Linear Alignment model [NLA, 3, 18], which assumes the three-dimensional intrinsic shear field $\tilde{\gamma}_{\alpha,I}$ should be proportional to the linear term of the gravitational tidal field s_{α} and weighted by the galaxy overdensity field $(1 + \delta_g)s_{\alpha}$.

We can therefore model the three-dimensional Intrinsic shear field by absorbing the weighting of the galaxy overdensity field into an amplitude constant,

$$\tilde{\gamma}_{\alpha,I} = -C \frac{\rho_{\text{crit}} \Omega_m}{D(z)} \left(\frac{1+z}{1+z_0} \right)^{\eta_1} s_{\alpha}. \quad (15)$$

There are two nuisance parameters to be marginalized in this model, the overall amplitude and the power index η_1 .

• Redshift and shear calibration

We employ a constant shift Δz in the redshift distribution to describe the uncertainty in redshift distribution $n(z)$,

$$n(z) \rightarrow n(z - \Delta z). \quad (16)$$

We marginalize one shift parameter per lens and source tomographic bin.

In weak lensing shape measurements, the bias introduced by a range of observational effects, such as PSF modeling, blending, and the under-sampling of the image, can be modeled as a multiplicative bias,

$$\gamma_{\alpha} \rightarrow (1 + m)\gamma_{\alpha}. \quad (17)$$

We marginalize one shear calibration bias parameter per source tomographic bin.

• Linear galaxy bias

We assume a linear relation between galaxy overdensity field and the matter overdensity field,

$$\delta_g = b_g \delta_m. \quad (18)$$

We assume one galaxy bias parameter per lens tomographic bin.

B. Approximations

1. Limber approximation

The angular spectrum calculation involves an integration of the spherical Bessel function, for example Eq. 6, which is highly oscillatory and numerically difficult to integrate. However, with the limitation of $x > \ell \gg 1$, the $j_\ell(x)$ contributes mostly at $x = \ell + \frac{1}{2}$, therefore, the spherical harmonic function can be approximated as a Dirac function,

$$j_\ell(k\chi) = \sqrt{\frac{\pi}{2\ell+1}} \delta_D(k\chi - \ell - \frac{1}{2}), \quad (19)$$

with the relation of Dirac function,

$$\begin{aligned} \delta_D(k\chi_1 - \ell - \frac{1}{2}) \delta_D(k\chi_2 - \ell - \frac{1}{2}) \\ = \frac{\delta_D(k - (\ell + 1/2)/\chi_1)}{\chi_1} \frac{\delta_D(\chi_1 - \chi_2)}{k}. \end{aligned} \quad (20)$$

Angular power spectrum thus has the general form under the Limber approximation,

$$C_{AB}^{ij}(\ell) = \int d\chi \frac{W_A^i(\chi) W_B^j(\chi)}{\chi^2} P_{AB}(k = \frac{\ell + 1/2}{\chi}, z(\chi)), \quad (21)$$

where A/B represent the observational field with P_{AB} the corresponding three-dimensional power spectrum.

However, there are a few physical limitations to this approximation: (1) it is applicable only when $\ell \gg 1$. (2) it requires that the weight function $W(\chi)$ is wide enough so that the integration can cover multipoles much greater than 1, i.e. $k\Delta\chi \gg 1$. Therefore, the assumption that $j_\ell(x)$'s oscillation parts cancel each other and that only $x \sim \ell + \frac{1}{2}$ mostly contributes to the integration is established. (3) the weight functions between two observables should largely overlap.

For cosmic shear and galaxy-galaxy lensing, Limber has been shown to be sufficiently accurate for surveys such as Rubin LSST-Y1 and DES-Y6 [13], as their galaxy sample distributions are wide and statistical power is much lower than that of the full-year of Stage-IV surveys. However, neither of the above approximations is proved to be hold for Roman HLIS-Y5 and Rubin LSST-Y10. In this work, we only implement non-Limber algorithm for galaxy clustering, which we explain in Section V A.

Beyond Limber calculations are typically supported via the algorithm Fast Fourier Transform in Logarithmic space (FFT-Log) [13]. Like Eq. 6, we need to solve an integration in a general form

$$\begin{aligned} \int d\chi_1 W^A(\chi_1) \int d\chi_2 W^B(\chi_2) \\ \cdot \int \frac{2k^2}{\pi} dk P^{AB}(k, z_1, z_2) j_\ell^{(n^A)}(k\chi(z_1)) j_\ell^{(n^B)}(k\chi(z_2)), \end{aligned} \quad (22)$$

where the P^{AB} is the power spectrum of fields A and B and it depends on the nonlinear matter power spectrum in some ways. The usual approach is to divide P^{AB} into two parts, linear part, the part depends on linear matter power spectrum P_{lin} , and nonlinear part, the part depends on nonlinear contribution of matter power spectrum $P_\delta - P_{\text{lin}}$. The nonlinear part can be directly computed by implementing the Limber approximation, because the Limber approximation works well at small scales, where the nonlinear contribution is significant. For the linear part we can decompose it into redshift-independent component and redshift-dependent transfer functions. This decomposition allows the term to be rewritten in terms of two one-dimensional single Bessel integrals, which can be computed efficiently using FFTLog.

2. Redshift space distortion

A second effect we consider is a correction to the projected galaxy overdensity field due to galaxy peculiar velocity with respect to the Hubble Flow, an effect referred to as redshift space distortion (RSD).

We follow the formalism of [15, 25]. Consider that the galaxy located at χ has a peculiar velocity along the line-of-sight $v_\parallel(\chi) = \mathbf{v}(\chi) \cdot \hat{\mathbf{n}}$ compared to the Hubble Flow, which introduces a redshift $1 + z_{\text{pec}} = 1 + v_\parallel(\chi)/c$, where we take non-relativistic approximations that $v_\parallel \ll c$, while the redshift induced by cosmic expansion is $1 + z_{\text{hub}} \equiv 1/a(\chi)$. Therefore, the observed redshift of the galaxy is $z_{\text{obs}} = (1 + z_{\text{pec}})(1 + z_{\text{hub}}) - 1 = z_{\text{hub}} + v_\parallel(\chi)/(a(\chi)c)$. This indicates that the real redshift distribution should have a correction in redshift, $n(z + \delta z)$ where $\delta z = v_\parallel/(a(\chi)c)$, or equivalently $W_{\delta,g}(\chi + \delta\chi)$, where $\delta\chi \approx \delta z/H(\chi)$, accounting for the impact of the peculiar velocity of galaxies.

With this correction, Eq. 1 becomes,

$$\begin{aligned} \delta_{g,\text{obs}}^i &= \int d\chi W_{\delta,g}^i(\chi + \delta\chi) (1 + \delta_g^{(3D)}(\hat{\mathbf{n}}\chi, \chi)) - 1 \\ &= \int d\chi (W_{\delta,g}^i + \frac{\partial W_{\delta,g}^i}{\partial \chi} \delta\chi) (1 + \delta_g^{(3D)}(\hat{\mathbf{n}}\chi, \chi)) - 1 \\ &\approx \delta_{g,D}^i(\hat{\mathbf{n}}) - \int d\chi W_{\delta,g}^i \frac{\partial}{\partial \chi} (\frac{\hat{\mathbf{n}} \cdot \mathbf{v}(\hat{\mathbf{n}}\chi, \chi)}{a(\chi)H(\chi)}) \\ &= \delta_{g,D}^i + \delta_{g,\text{RSD}}^i. \end{aligned} \quad (23)$$

In the second line of Eq. 23, we regard $\delta\chi$ as a small perturbation and expand the normalized selection function by the Taylor Expansion. We separate the second term of the last line of Eq. 23 by $\delta_{g,\text{RSD}}$, denoting the linear contribution of RSD to the galaxy overdensity field.

The derivative in the third line of Eq. 23 can be approximated as $f\mu^2\delta_m$ in harmonic space under the linear approximation, where μ is defined as $k_\parallel/k$, while the f is the growth factor.

3. Nonlinear matter power spectrum

An accurate nonlinear matter power spectrum is critical to recover the true cosmology in the 3x2pt inference pipeline,

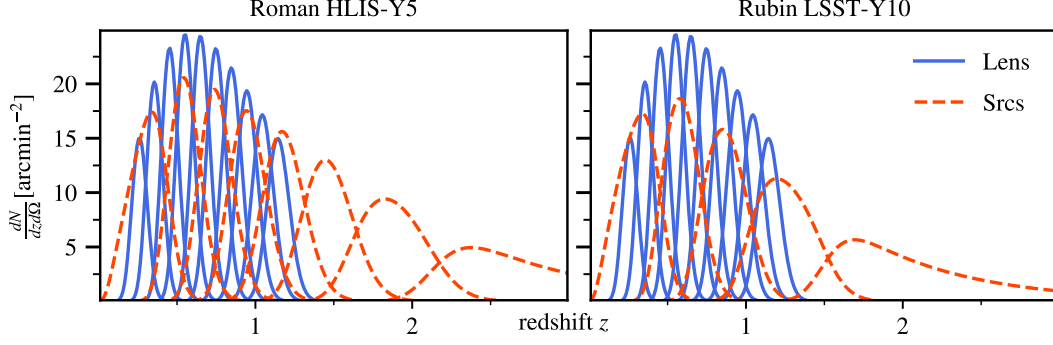


FIG. 1. These plots show redshift distribution of galaxy samples for Roman (left) and Rubin (right) used in this work, as defined in Section III A and III B.

as it describes the matter distribution to smaller scales. In most state-of-the-art 3×2pt analyses [9, 17], HMCode [23] has been the model used. However, recent developments with the EUCLIDEMULATOR2 [11] has shown that better, higher-resolution N-body simulations can result in more accurate predictions for the nonlinear power spectrum. As such, we use the EUCLIDEMULATOR2 as our baseline nonlinear matter power spectrum model, and test the impact on using HMCode as an alternative model. We describe briefly below each of these implementations.

• EUCLIDEMULATOR2

EUCLIDEMULATOR2 provides precise predictions of the non-linear matter power spectrum inside an eight-dimensional cosmological parameter space, including: Λ CDM cosmological parameters ($\Omega_b, \Omega_m, A_s, n_s, h$), dark energy equation of state parameters (w_0, w_a), and the sum of neutrino mass $\sum m_\nu$. EUCLIDEMULATOR2 is a regression model, optimizing parameters of polynomial chaos expansion (PCE) to predict the weights of 14 PCA bases of the matter power spectrum from cosmological parameters. The actual training is done with a dataset including 127 high-resolution simulations based on a pre-trained model, whose training dataset is from HALOFIT [33].

The final EUCLIDEMULATOR2 predictions achieves 1% accuracy for the non-linear correction factors under $z \leq 3$ and $0.01 h\text{Mpc}^{-1} \leq k \leq 10 h\text{Mpc}^{-1}$, and it is so far the most precise non-linear matter power spectrum model among those widely used by the weak lensing community. Since EUCLIDEMULATOR2 has a strict boundary for k -modes, we calculate k -modes that exceed $10 h\text{Mpc}^{-1}$ with HMCode for consistency. The EUCLIDEMULATOR2 has a strict boundary for cosmological parameters, so we follow that to set the cosmological parameter priors as shown in Table III.

• HMCode

We choose HMCode as an alternative model of the non-linear matter power spectrum, as it provides analytic and precise reconstructions, with an RMS error of less

than 2.5 percent compared to simulated data over a wide range of cosmologies, scales $k < 10 h\text{Mpc}^{-1}$, and redshifts $z < 2$. HMCode is reliable in non-standard cosmologies since it predicts the non-linear matter power spectrum analytically. It also provides a parameterization prescription to model the impact of baryonic feedback. However, we do not use this feature as our scale cut should eliminate the majority of the contribution from baryons.

HMCode is based on the Halo model [31] and modifies it with a series of parameterizations to match the actual power spectrum from simulations. The parameters include mass function, concentration parameters, variables that describe properties of the linear spectrum, a spectral index to smooth the transition between the one- and two-halo terms, etc. The parameterization is independent of cosmological parameters, enabling maximum applicability to novel cosmological scenarios, while parameters related to non-linear matter power spectrum are calibrated by 37 representative simulations.

III. SURVEY SPECIFICATION

In this section we briefly describe the baseline design and data characteristics that will go into our forecasting exercise for Roman (Section III A) and Rubin (Section III B). The goal is to model the source and lens samples for a “fiducial” 3×2pt analysis for the *final* Roman and Rubin dataset, assuming 5 years for Roman HLIS data (Roman HLIS-Y5) and 10 years for Rubin data (Rubin LSST-Y10).

For Rubin we will mostly follow the LSST Dark Energy Science Collaboration (DESC) Science Requirement Document (SRD) [8]. For Roman, there has been a number of proposed source and lens samples in the literature with different analysis choices [5, 10]. Without loss of generality, here we opt to use the source sample defined in the Data Challenge 1 (DC1, manuscript in preparation) of the Cosmological Parameters Inference Pipeline (CPIP) group of Roman and the same lens sample used in the Rubin case. We will show in Appendix C

the impact of using the different samples.

The final redshift distributions used for the two surveys in this paper is shown in Figure 1 and the survey specifications are summarized in Table I.

A. Roman

The 3×2pt analysis in Roman will use the imaging data from Roman’s High Latitude Wide Area Survey (HLWAS), referred to as High Latitude Imaging Survey (HLIS). In the most recent design [39], HLIS will conduct three tiers of surveys: medium, deep, and wide. We will focus on the medium tier, with a characteristic imaging depth of $m_{AB} = 26.5$ in three imaging bands (Y,J,H) for a 5σ point-source, covers $2,415 \text{ deg}^2$ with an effective source galaxy number density of $n_{\text{eff}}^s \approx 41 \text{ arcmin}^{-2}$. We carry out our analysis using only the medium tier because it is uncertain that we will be able to properly characterize the photometric redshift in the wide tier and the deep tier is too small to carry out a proper cosmological analysis.

Following DC1, we generate the source redshift distribution that the Galaxy Survey Exposure Time Calculator⁴ (ETC). This ETC is first employed to the CANDELS catalog [16] with the same S/N cuts as [10] to get the distribution of detected samples. We then fit the shape, split it into 8 redshift bins with an equivalent number density per bin, and convolved bins with Gaussian distribution with width of 0.05.

B. Rubin

The 3×2pt analysis from Rubin will use the Wide-Fast-Deep (WFD) component from the Legacy Survey of Space and Time (LSST). LSST is estimated to achieve 5σ point-source detection depths of 25.30, 26.84, 27.04, 26.35, 25.22, 24.47 in (u, g, r, i, z, y) over $14,300 \text{ deg}^2$ of sky. It is expected to provide an effective number density of $n_{\text{eff}}^l \approx 48 \text{ arcmin}^{-2}$ for lens samples and $n_{\text{eff}}^s \approx 27 \text{ arcmin}^{-2}$ for source samples.

To describe the redshift distributions for the galaxy samples, we use the analytical smail model [32] to specify the distributions, as done in the LSST DESC SRD,

$$\frac{dN}{dz} \propto z^2 \exp[-(z/z_0)]^\alpha. \quad (24)$$

For the lens sample, we set $z_0 = 0.28, \alpha = 0.90$ in Eq. 24 and require the total number density to equal 48 arcmin^{-2} . The fitted redshift distribution is then uniformly divided into 10 tomographic bins between redshift $0.2 \leq z \leq 1.2$, therefore, some samples at the lowest and highest redshifts are lost and the rest effective number density is about 30 arcmin^{-2} .

For the source sample, we set $z_0 = 0.11, \alpha = 0.68$ and require the total number density to equal 27 arcmin^{-2} . The fitted redshift distribution is then divided into 5 tomographic bins with equivalent number density.

TABLE I. Survey settings for Roman HLIS-Y5 and Rubin LSST-Y10. Survey area is assumed medium tier for Roman HLIS-Y5. We assume both surveys use the Rubin LSST-Y10 lens samples, and the total number density is 30 arcmin^{-2} after tomographic binning.

	Roman HLIS-Y5	Rubin LSST-Y10
Survey area (deg^2)	2,415	14,300
Lens number density n_{eff}	48 (30)	48 (30)
SrCs number density n_{eff}	41.00	27.73
Shape noise σ_e	0.26	0.26
Number of lens bins	10	10
Number of srCs bins	8	5
Number of θ bins	20	20
θ range (arcmin)	(1, 500)	(1, 500)
Number of ℓ bins	20	20
ℓ range	(20, 4000)	(20, 4000)

IV. SIMULATED LIKELIHOOD ANALYSIS

In this section we describe our simulated likelihood analysis. We will primarily use the COBAYA-CosmoLike Joint Architecture⁵ (Cocoa) for this analysis. Cocoa is built upon the cosmology library CosmoLike [19] to simulate the 3×2pt data vectors and employs the COde for BAYesian Analysis framework [COBAYA, 36] to perform the inference. Besides the default MCMC sampler of COBAYA, Cocoa also integrates a series of sampler tools, including POLYCHORD, NAUTILUS and EMCEE.

It is worth noting that Cocoa also integrates a wide range of analysis tools which are intensively used in the cosmological inference, such as likelihood minimizer, likelihood profile, and tension metrics. And it will provide a set of emulators emulating commonly used inference likelihood and CAMB to accelerate the sampling process. One of the main features of Cocoa is that it introduces shell scripts that allow users to containerize COBAYA, the Boltzmann codes, and multiple likelihoods, ensuring that users can adopt consistent versions for the Fortran/C/C++ compilers and libraries across multiple machines.

We will use the Core Cosmology Library [CCL, 7] as a reference pipeline, it is a public standardized library of routines to calculate basic observables which will be the standard theoretical prediction package for LSST Dark Energy Science Collaboration (DESC). In Section V A, we will compare our model and CCL at the data vector level, ensuring that no major modeling mismatch exists.

In this work we will perform simulated likelihood analysis with “baseline” configurations for Roman HLIS-Y5 and Rubin LSST-Y10 as described in Section III. We will examine how the Λ CDM constraints change when we employ the Limber approximation, turn off RSD and change our modeling for the nonlinear matter power spectrum. We will also examine these results in both real and harmonic space. To perform this study we first describe in Section IV A our procedures in determining the scale cuts for the different 3×2pt data vectors.

⁴ <https://roman.gsfc.nasa.gov/science/etc14.html>

⁵ <https://github.com/CosmoLike/cocoa>

We next describe in Section IV B our inference framework and what we choose to use for the various nuisance parameters.

A. Scale cut

Unmodeled components at small scales are one of the main sources of theoretical uncertainties. For cosmic shear, the main uncertainty comes from our (lack of) understanding of the distribution of baryons on small scales due to the feedback [6, 27]. For two-point functions involving galaxy density, the main uncertainty comes from nonlinear galaxy bias.

There are a wide range of approaches in the literature that attempt to model these small scales [1, 30]. In general, although recent results have shown promise in constraining these small-scale effects with external data [37, 38], the community as a whole has not converged on an agreed-upon model and a set of agreed-upon priors. As such, in this analysis we will use a slightly conservative approach that is well-validated by Stage-III surveys – we will remove the small scales that are uncertain in our analysis.

We note that there has been previous attempts to perform Roman and Rubin forecasts that have assumed a different approach towards scale cuts. For example, the LSST SRD has assumed $\ell_{\max} = 3000$ for cosmic shear, and $k_{\max} = 0.3h/\text{Mpc}$ or equivalently $R_{\min} = 21\text{Mpc}/h$ for galaxy-galaxy lensing and galaxy clustering. In [5, 10], they used the same assumptions as the LSST SRD for Roman forecasts. In this work, we will instead be using an approach more similar to what is done in the DES-Y6 cosmology analysis [9]. We compare the constraining power of different scale cuts in Appendix C.

We will define the scale cuts in real and harmonic spaces and how to convert scale cuts from real space to harmonic space. For cosmic shear, we set a $\Delta\chi^2$ budget for each tomographic bin pair and thus iteratively determine the scale cut by requiring the $\Delta\chi^2$ between contaminated and fiducial data vectors to fall below the budget. Specifically, we generated baryonic feedback to cosmic shear from ILLUSTRIS simulations [24], and set $\Delta\chi^2 = 0.025$ budget per bin in real space, while the number is 0.05 for that in harmonic space to balance the total $\Delta\chi^2$ in both spaces. For galaxy-galaxy lensing and galaxy clustering, we set a physical minimum transverse radius $(R_{\min}^{\gamma t}, R_{\min}^w) = (6\text{Mpc}/h, 8\text{Mpc}/h)$. We take the approximation $\theta_{\min} = R_{\min}/\chi(\bar{z})$ to convert the minimum transverse radius into the minimum angle in the real space, where $\bar{z}$ is the effective average redshift of a lens bin. We then utilize the relation $\ell_{\max} = \pi/\theta_{\min}$ to convert the minimum angle to the largest multipole, i.e. the scale cut in harmonic space. For both Roman and Rubin, we measure 20 angular bins in the angular range $1' < \theta < 500'$ in real space while measuring 20 multipoles in the range $20 < \ell < 4000$ in harmonic space. Table II shows the signal-to-noise (S/N) values for each part of the data vector after scale cuts.

TABLE II. The Signal to Noise Ratio (SNR) or the $\sqrt{\Delta\chi^2}$ for each probe after our fiducial scale cuts. The $\Delta\chi^2$ is defined as $\Delta\chi^2 = \mathbf{D}^T \mathbf{Cov}^{-1} \mathbf{D}$. The inconsistency of SNRs in both spaces is due to the scale cut.

Probe	Roman HLIS-Y5	Rubin LSST-Y10
$\xi^{\pm}$	72	101
γ_t	157	340
$w(\theta)$	122	297
total	191	427
C_{ℓ}^{ss}	46	64
C_{ℓ}^{gs}	180	389
C_{ℓ}^{gg}	302	738
total	332	791

TABLE III. Model parameters for the analysis. Superscript i refers to the indice of the tomographic bin. $\mathcal{U}(a, b)$ represents a uniform distribution between a and b , while $\mathcal{N}(\mu, \sigma)$ represents a gaussian distribution centered at μ with a variance σ^2 .

Parameter	Fiducial	Prior
$A_s \times 10^9$	2.1	$\mathcal{U}(1.7, 2.5)$
n_s	0.96605	$\mathcal{U}(0.92, 1.00)$
H_0	67.32	$\mathcal{U}(61, 73)$
Ω_b	0.0495	$\mathcal{U}(0.04, 0.06)$
Ω_m	0.316	$\mathcal{U}(0.24, 0.40)$
w	-1.0	$\mathcal{U}(-1.3, -0.7)$
Lens photo-z Δz^i	$\{0\}^i$	$\mathcal{N}(0, 0.003)$
Srsc photo-z Δz^i	$\{0\}^i$	$\mathcal{N}(0, 0.001)$
Source calibration m^i	$\{0\}^i$	$\mathcal{N}(0, 0.003)$
Intrinsic Alignment A_{1A}, α_{1A}	0.5, 0	$\mathcal{U}(-5, 5), \mathcal{U}(-5, 5)$
Galaxy Bias b^i	$\frac{0.95}{D(\bar{z}^i)}$	$\mathcal{U}(0.8, 2)$

B. Covariance matrix and inference

We use COCOA to perform the inference of parameters $\mathbf{p}$ from the observed 3×2 pt data $\mathbf{D}$. Throughout we assume a Gaussian likelihood of the form,

$$\mathcal{L}(\mathbf{D}|\mathbf{p}) \propto \exp\left(-\frac{1}{2} [\mathbf{D} - \mathbf{M}(\mathbf{p})]^T \mathbf{Cov}^{-1} [\mathbf{D} - \mathbf{M}(\mathbf{p})]\right), \quad (25)$$

where $\mathbf{M}(\mathbf{p})$ is the model for the observables, and $\mathbf{Cov}$ is the covariance matrix for the data. We now describe the model, its parameters, their priors, and the covariance matrix, and then the tools of the inference process.

• Model

$\mathbf{M}(\mathbf{p})$ comprises three power spectrum of probes, cosmic shear C_{ℓ}^{ss} , galaxy-galaxy lensing C_{ℓ}^{gs} and galaxy clustering C_{ℓ}^{gg} as described in Section II A. Then they are converted to the correlation functions, $(\xi^+, \xi^-, \gamma_t, w(\theta))$ in the real space by the Eq. 8.

• Parameters and priors

To simulate the data vector $\mathbf{M}(\mathbf{p})$, cosmological and nuisance parameters $\mathbf{p}$ are required. In this paper, we sample and marginalize over five Λ CDM cosmological

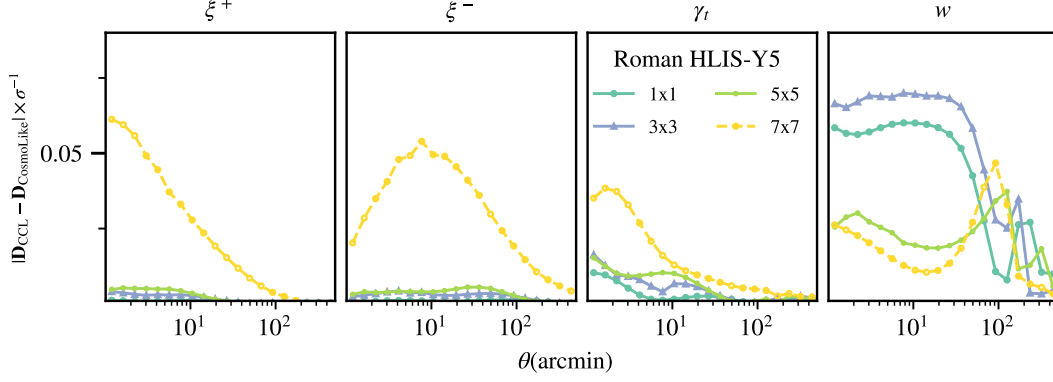


FIG. 2. This figure shows the bias significance of data vector between two pipelines over the angular scale: CCL and CoCoA. Data vector difference is first computed and then divided by the square root of diagonal term of data vector covariance matrix to get the significance of bias. Each of the four subplots represents a probe, including $(\xi^+, \xi^-, \gamma_t, w)$. Only part of the correlations are shown in the figure, for example, (1,1) represents the auto-correlation of the first source bin for cosmic shear, or the correlation between the first lens bin and the first source bin for galaxy-galaxy lensing, or the auto-correlation of the first lens bin for galaxy clustering.

parameters $(A_s, n_s, H_0, \Omega_b, \Omega_m)$ as described in Section II A, while setting other cosmological parameters fixed (sum of neutrino mass m_ν , the equation of state parameter of dark energy w , and the reionization optical depth τ).

For astrophysical nuisance parameters, we set broad, flat priors since there is still large uncertainties in our understanding. We set the amplitude and power index of the NLA intrinsic alignment model to $\mathcal{U}(-5, 5)$, and the linear galaxy bias parameters to $\mathcal{U}(0.8, 2)$.

For observational nuisance parameters associated with redshift and shear calibration, we set strong Gaussian priors given the advance in Stage-III surveys – most of these effects are now understood very precisely. We use Gaussian priors $\mathcal{N}(\sigma = 0.003)$ for the lens galaxy redshift uncertainty and $\mathcal{N}(\sigma = 0.001)$ for the source galaxy redshift uncertainty. We use $\mathcal{N}(\sigma = 0.003)$ for shear calibration uncertainty. These numbers are based on the LSST DESC SRD but also is assumed to be similar for Roman.

The fiducial values and priors of these parameters are described in Table III.

• Covariance matrix

We analytically simulate the covariance matrix using the package CosmoCov [12]. It analyzes the covariance matrix into three parts, the Gaussian covariance, non-Gaussian covariance, and the super-sample covariance as

$$\text{Cov}(C(\ell_1), C(\ell_2)) = \text{Cov}^G(C(\ell_1), C(\ell_2)) + \text{Cov}^{\text{NG},0}(C(\ell_1), C(\ell_2)) + \text{Cov}^{\text{SSC}}(C(\ell_1), C(\ell_2)).$$

The equations calculating the covariance of three probes includes: cosmic shear, galaxy-galaxy lensing, and galaxy clustering can be found in [20]. While the transform of covariances from harmonic space to real

space follows [13], which employs the 2D – FFTLog algorithm to accelerate the double Bessel integration involved in this transform.

• Sampling algorithm

We employ the Metropolis sampler used by CosmoMC [22] throughout this paper. We assess convergence of chains by Gelman-Rubin statistics $\hat{R}$ and stop sampling when $\hat{R} - 1 < 0.05$ for all marginalizing parameters and $\hat{R} - 1 < 0.2$ for all 95% credible intervals.

V. RESULTS

In this section we present the main results of this study. We first perform a comprehensive validation check of the CosmoLIKE model for the 3×2pt probes in Section V A, by comparing it with models generated by CCL. Having validated the model, we define in Section V B our numerical experiment setup, and next in Section V C we use CoCoA (which wraps around CosmoLIKE) to quantify the impact of Limber approximation, RSD and choices of nonlinear matter spectrum on the 3×2pt data vector and the resulting cosmological constraints for both Roman HLIS-Y5 and Rubin LSST-Y10.

A. Code comparison: CosmoLIKE vs. CCL

Given that CoCoA is a relatively new code base, we like to validate the modeling aspect of CoCoA, or CosmoLIKE, to existing code. To this end, we compare the data vectors generated from CosmoLIKE with that from CCL under the fiducial Roman configuration (first column in Table I). We first compare all the building blocks used to compute the 3×2pt data vectors: the matter power spectrum, the comoving radial distance, the radial kernel, and the lensing efficiency. Next we combine the components together and compare the final 3×2pt data vector.

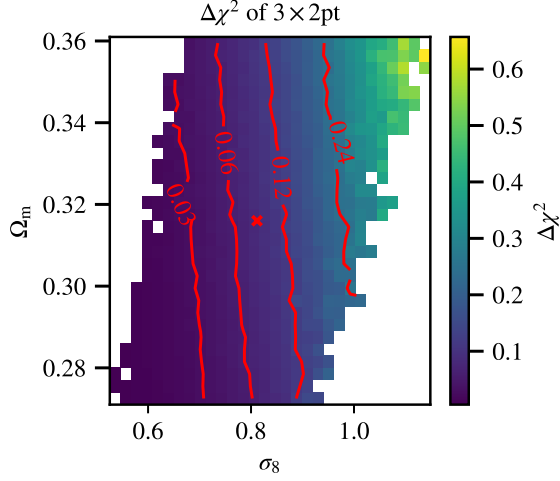


FIG. 3. $\Delta\chi^2$ of $3 \times 2\text{pt}$ with varying cosmologies over $\Omega_m - \sigma_8$, which are simulated by CosmoLike and CCL assuming our fiducial cosmology. The red lines with numbers are $\Delta\chi^2$ contours over cosmologies. The red cross represents our fiducial cosmology.

We also investigate whether the two pipelines agree under different assumptions of systematic effects (varying parameters in the IA model) and approximations (RSD and non-Limber) that are used in this analysis.

For the real-space data vector with and without fiducial scale cuts, we find $\Delta\chi^2_{\text{fid-cut}}(\xi^\pm, \gamma_t, w, 3 \times 2\text{pt})_{\text{Roman}} = (0.004, 0.007, 0.052, 0.137)$ and $\Delta\chi^2(\xi^\pm, \gamma_t, w, 3 \times 2\text{pt})_{\text{Roman}} = (0.032, 0.052, 0.064, 0.167)$. For the harmonic-space data vector with and without scale cuts, we find $\Delta\chi^2_{\text{fid-cut}}(C^{ss}, C^{gs}, C^{gg}, 3 \times 2\text{pt})_{\text{Roman}} = (0.003, 0.005, 0.031, 0.075)$ and $\Delta\chi^2(C^{ss}, C^{gs}, C^{gg}, 3 \times 2\text{pt})_{\text{Roman}} = (0.026, 0.062, 0.050, 0.198)$. These values are all small (mostly $\ll 1$, especially when scale cuts are applied), demonstrating that two modeling pipelines agree well with each other in our fiducial settings. In Figure 2 we show one example for the difference in part of the real-space data vector in the Roman baseline setting.

There are a few subtleties in this comparison that we like to highlight, which are important for future similar comparisons

- In all these tests, we import the nonlinear power spectrum of CosmoLike into CCL using its Pk2D module, thus two pipelines use the exactly same power spectrum. We do this since both codes wrap around CAMB and the exact implementation of each of the codes is not of interest in this work.
- When we turn off RSD, the $\Delta\chi^2(3 \times 2\text{pt})$ increases significantly to ~ 1 and the scale cut does not mitigate this effect. This is due to a minor inconsistency in the implementation of the non-Limber algorithm in galaxy clustering, which becomes negligible once RSD is turned on. As $\Delta\chi^2 \sim 1$ is still much smaller than the effects of interest that we will discuss in Section V C, we do not attempt to correct for it. But future work may need to reconcile this difference.

- At the time of this analysis, in CosmoLike, non-Limber is only implemented for galaxy clustering, where the main effect lies, and not for galaxy-galaxy lensing or cosmic shear. We can estimate the effect of this approximation using CCL. We find $\Delta\chi^2(\xi^\pm, \gamma_t)_{\text{Roman}} = (0.30, 0.21)$ for Roman HLIS-Y5 and $\Delta\chi^2(\xi^\pm, \gamma_t)_{\text{Rubin}} = (1.16, 0.56)$ for Rubin LSST-Y10. We determine that this level of discrepancy is small enough for us to continue our analysis with the existing CosmoLike.
- For numerical convergence reasons, the implementation of the non-Limber power spectra requires one to set a maximum multiple $\ell_{\text{max}}^{\text{NonLimber}}$ for the non-Limber calculation. This method is used for both CosmoLike and CCL. For multipoles beyond $\ell_{\text{max}}^{\text{NonLimber}}$, the calculation defaults back to using the Limber approximation. We will set $\ell_{\text{max}}^{\text{NonLimber}} = 150$ when calculating the non-Limber power spectrum from both CosmoLike and CCL throughout this analysis. The introduction of $\ell_{\text{max}}^{\text{NonLimber}}$ results in a slight underestimate of difference in both data vector and constraints when comparing Limber vs. non-Limber later in Section V C.
- Our code-comparison exercise revealed two effects in the real space data vectors that needed additional caution: the angular bin window function (or, bin-average) and the flat/curve-sky transformation. We describe our tests in Appendix A. Both effects are correctly implemented in CCoA, so here we mostly point out the impact of the results if the effects are not accounted for.

Next, we calculate the data vector $\Delta\chi^2$ for a range of cosmological parameters to check for potential cosmology-dependence effects. Figure 3 shows the $\Delta\chi^2$ for $3 \times 2\text{pt}$ as a function of different Ω_m and σ_8 values. We find that the $\Delta\chi^2$ can rise to around 0.6 when Ω_m becomes larger, though this number is still sufficiently small. We argue that such dependence on cosmology is mainly due to the overall amplitude increase as the Ω_m rises, illustrating that we should take this cosmology-enhanced $\Delta\chi^2$ into consideration when we compare data vectors at cosmology that has an overall smaller amplitude of signal.

B. Experiment setup

We like to investigate the three different modeling approximations described in Section II B: Limber approximation made for matter power spectrum, RSD as a correction to the observed galaxy overdensity field, and different models used for the nonlinear matter power spectrum. For Limber and non-Limber, we only investigate its effect for galaxy clustering in this paper as the Limber approximation does not impact the galaxy-galaxy lensing and cosmic shear data vectors significantly. For RSD, we investigate its influence on galaxy-galaxy lensing and galaxy clustering. We test the impact of the nonlinear matter power spectrum for the full $3 \times 2\text{pt}$ data vector.

To compare these approximations systematically, we build five model setups.

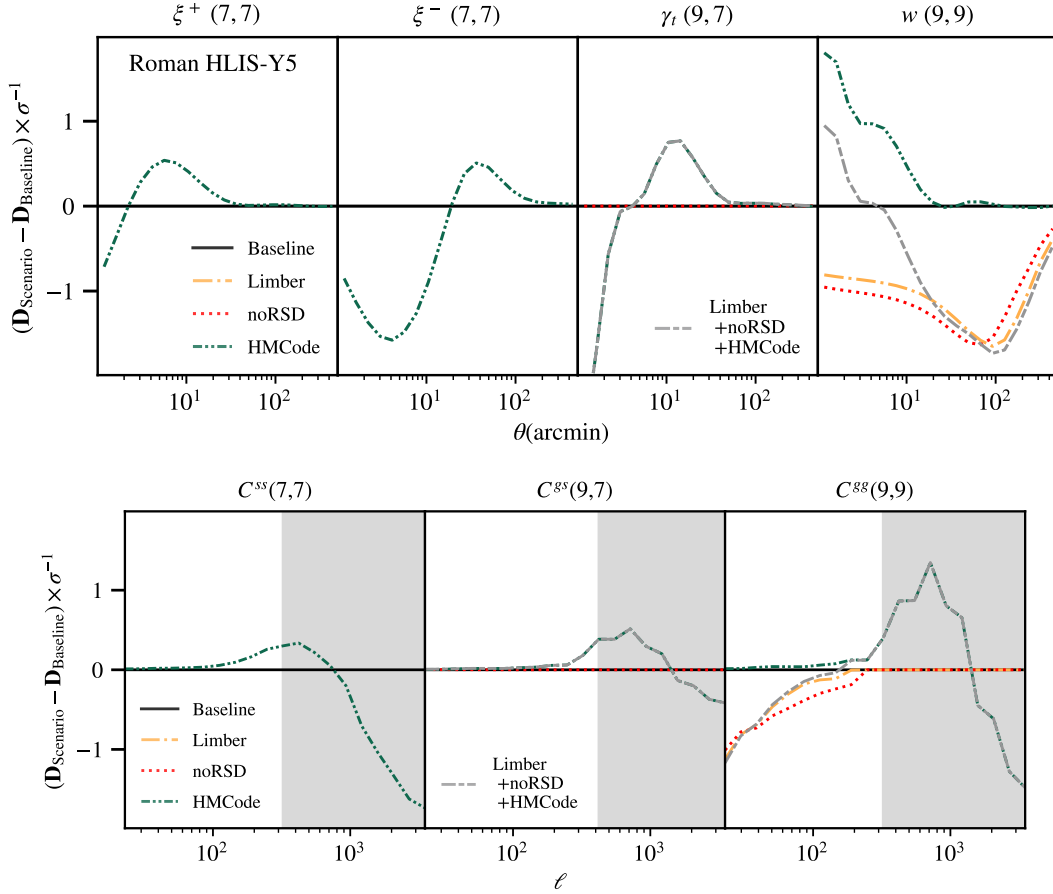


FIG. 4. This plot compares data vectors of four scenarios to our Baseline, including Limber, noRSD, HMCode, and Limber-noRSD-HMCode. The data vector difference is measured by the diagonal components of data vector covariance matrix as $(\text{Scenario} - \text{Baseline})/\text{Diag}(\text{Cov})$, illustrating the significance of bias. The upper figure is for Roman HLIS-Y5 in the real space while the lower figure is for that in harmonic space. The gray shadow represents the DES-like scale cut. All showed combinations are using the lens and source galaxy sample with the highest redshift as the influence of these approximation is more significant at these bins. Note that Limber, noRSD and Limber-noRSD-HMCode are not shown for cosmic shear, since the former two do not have an effect on this probe while the latter is the same as HMCode. Limber is not shown for galaxy-galaxy lensing for the same reason.

- **Baseline:** this is designed to be the most “correct” model setup, where we implement non-Limber for galaxy clustering, RSD for observed galaxy overdensity field, and EUCLIDEMULATOR2 for nonlinear matter power spectrum.
- **Limber:** this setup is designed to test the impact of Limber approximation, so it follows Baseline but uses Limber approximation for galaxy clustering.
- **noRSD:** this setup is designed to test the impact of ignoring RSD, so it follows Baseline but does not include the RSD correction in the galaxy overdensity field.
- **HMCode:** this setup is designed to test the impact of using a less precise nonlinear matter power spectrum, so it follows Baseline but uses HMCode for the matter power spectrum.
- **Limber-noRSD-HMCode:** this setup is designed to test

the impact of all three approximation combined, so it uses Limber approximation, no RSD and HMCode.

By comparing Baseline and other four setups respectively, we are able to quantify how these approximations affect the data vector simulations and cosmological inference.

C. Impact of approximations

We first simulate data vectors of these setups and quantify the discrepancy those approximation could have on the data vectors. All data vectors are simulated under four scenarios as for two surveys Roman HLIS-Y5 and Rubin LSST-Y10 (Table I), and for real and harmonic space. These scenarios are noted as Roman-Real, Roman-Harmonic, Rubin-Real, Rubin-Harmonic.

Figure 4 shows comparison for Roman-Real, Roman-Harmonic for a subset of the data vector, while the Figure 5

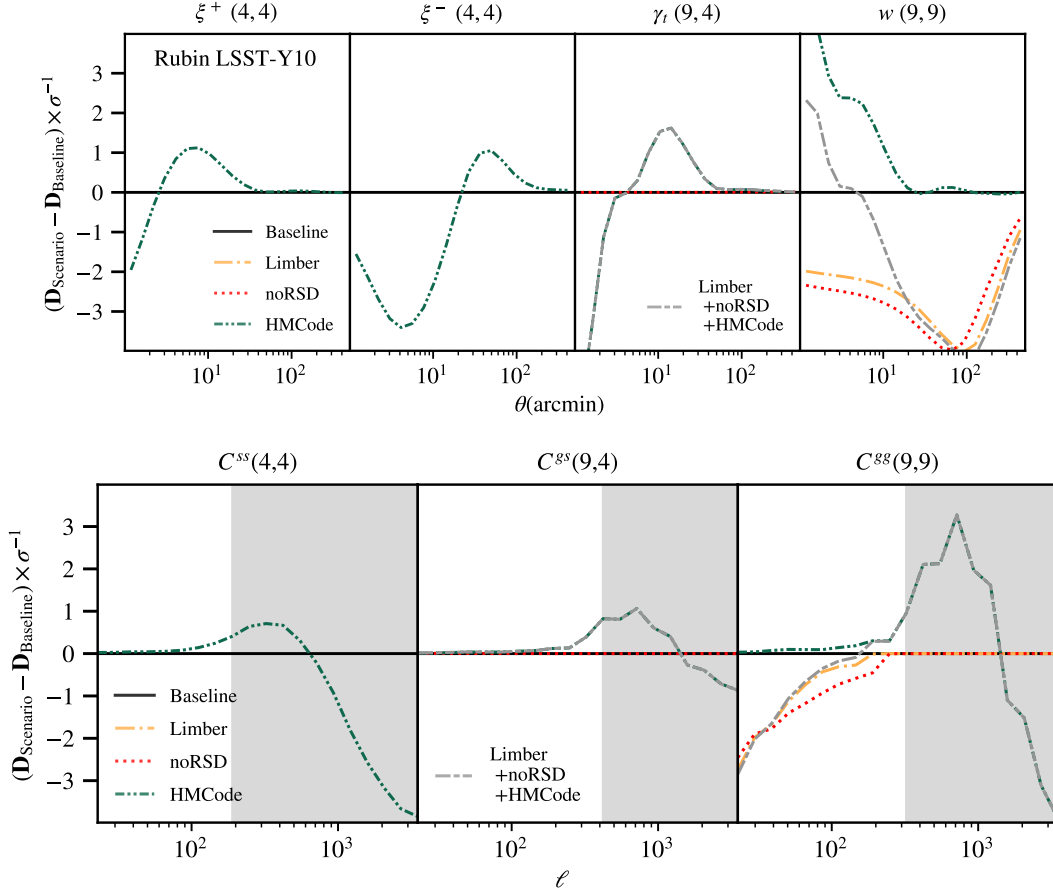


FIG. 5. This plot compares data vectors of four scenarios as described in Figure 4. The data vector difference is measured by the diagonal components of data vector covariance matrix as $(\text{Scenario} - \text{Baseline})/\text{Diag}(\text{Cov})$, illustrating the significance of bias. The upper figure is for Rubin LSST-Y10 in the real space while the lower figure is for that in harmonic space. The gray shadow represents the DES-like scale cut. All showed combinations are using the lens and source galaxy sample with the highest redshift as the influence of these approximation is more significant at these bins.

TABLE IV. $\Delta\chi^2$ between data vectors of different scenarios and the baseline, and the significance of bias in the (Ω_m, S_8) plane relative to the baseline chain.

	Limber	noRSD	HMCCode	Limber +noRSD+HMCCode
$\Delta\chi^2$				
DES-Y3	9.15	8.90	0.93	12.29
Roman-Real	23.85	23.66	6.75	34.79
Roman-Harmonic	13.43	16.33	1.13	15.00
Rubin-Real	131.65	129.10	34.99	190.62
Rubin-Harmonic	71.42	87.09	6.03	79.44
Significance of bias (σ)				
Roman-Real	0.73	0.70	0.61	0.96
Roman-Harmonic	0.75	0.75	0.56	1.12
Rubin-Real	1.79	2.05	0.98	2.76
Rubin-Harmonic	1.97	2.31	1.24	3.12

743 shows the same for Rubin-Real and Rubin-Harmonic. The

744 x-axis demonstrates the scale of observables which is either
 745 in angular scale (arcmin) or harmonic modes (ℓ), while the
 746 y-axis shows the significance of the data vectors' bias against
 747 **Baseline**, calculated by the difference dividing the diagonal
 748 components of data vector covariance matrix. For brevity, we
 749 only show one bin combination for each of the data vector type
 750 ($\xi_{\pm}$, γ_t , w and C^{ss} , C^{gs} , C^{gg}) – we choose to show the highest
 751 redshift bin as an illustration. Because Limber and noRSD
 752 demonstrates greater discrepancy to **Baseline** at higher red-
 753 shift. For HMCCode, although it shows a larger difference at
 754 lower redshift on small scales, the difference in intermediate
 755 scales are somewhat redshift-independent.

756 We first look at the cosmic shear panels. Since RSD only
 757 has influence for galaxy-galaxy lensing and galaxy clustering,
 758 and non-Limber is only applied to galaxy clustering, the only
 759 approximation that can affect cosmic shear is the nonlinear
 760 matter power spectrum simulator. As seen in Figure 4 and
 761 Figure 5, setup HMCCode and **Baseline** show a noticeable dif-
 762 ference when $\theta < 60'$ for ξ^+ , $\theta < 110'$ for ξ^- and $\ell > 100$ for
 763 C_{ℓ}^{ss} , though most of the affected scales are cut by our fiducial

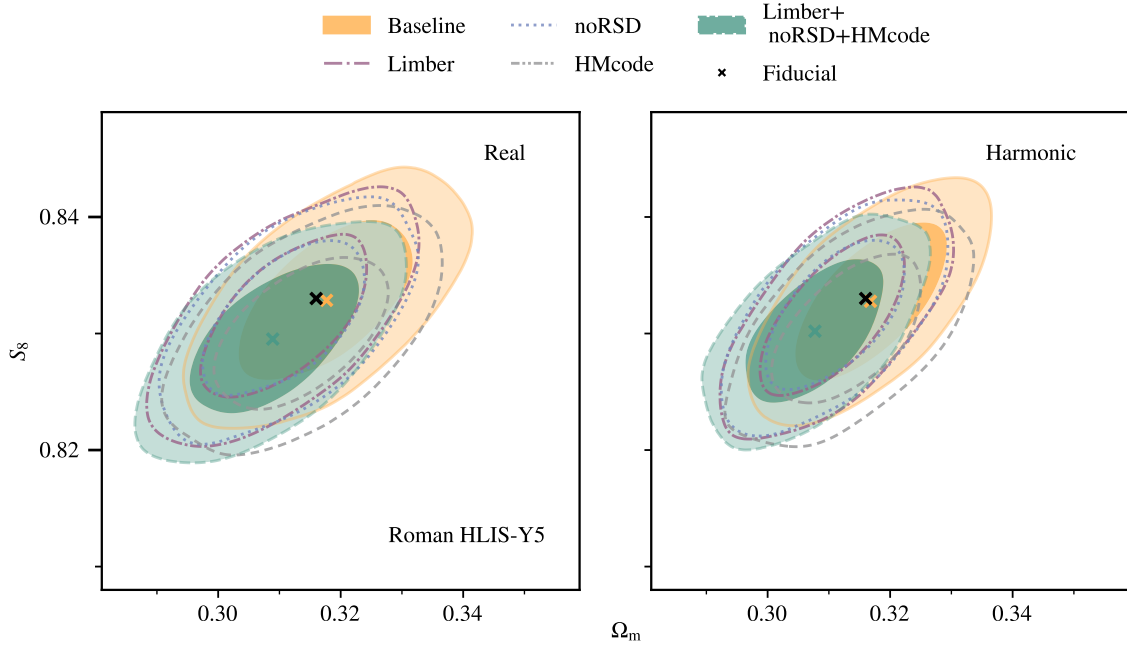


FIG. 6. This plot presents $S_8 - \Omega_m$ constraints of five scenarios, including Baseline, Limber, noRSD, HMcode, and Limber-noRSD-HMcode, for survey Roman HLIS-Y5. The inner and outer contours enclose the area representing 95% and 68% confidence level. The yellow and green cross represents the posterior means of scenario Baseline and the most biased scenario Limber-noRSD-HMcode. The black cross represents the input fiducial value.

scale cut. Its typical significance of bias to data point introduced by the nonlinear matter power spectrum is within 1σ for Roman HLIS-Y5 and about 2σ for Rubin LSST-Y10, while both of them are well controlled under 1σ after scale cut.

Next we look at the galaxy-galaxy lensing panels in Figure 4 and Figure 5. We see that the nonlinear matter power spectrum has the most significant impact on the data vector while that of RSD is negligible. This is because RSD is greatly suppressed under Limber approximation as its $k_{\parallel}$ mode is near 0 defined in Section II B 2. The influence of the nonlinear matter power spectrum becomes noticeable when $\theta < 60'$ and $\ell > 100$, and the typical bias significance to data point is below than 1σ for Roman HLIS-Y5 and about 1σ for Rubin LSST-Y10 after the fiducial scale cut.

Finally we examine the panels for galaxy clustering. The impact of the nonlinear matter power spectrum is still significant when $\theta < 20'$ and $\ell > 100$, which introduces a typical bias below 1σ for Roman HLIS-Y5 and around 1σ for Rubin LSST-Y10 after scale cut. The RSD is important as it introduces a typical 1σ bias to data vectors for Roman HLIS-Y5 and 2σ for Rubin LSST-Y10. Limber approximation has a similar level of impact as the RSD, but since by definition there is little RSD in the presence of Limber approximation, this demonstrates that RSD is the dominant effect between the two.

We show the resulting posteriors in the $\Omega_m - S_8$ plane in Figures 6 and 7. These plots allow us to translate the $\Delta\chi^2$ numbers in Table IV to shifts in cosmological constraints. We summarize the bias of posterior mean between Baseline and other setups in Table IV and discuss them below.

First, we show that in all scenarios, the posterior mean of the Baseline setup lies close to the input fiducial value as demonstrated by the orange cross (Baseline) and black cross (Fiducial) in Figure 6 and 7. Numerically, the bias significance of Baseline posterior mean from fiducial value is 0.25σ in Roman-Real, 0.18σ in Roman-Harmonic, 0.07σ in Rubin-Real, and 0.11σ in Rubin-Harmonic. These numbers mainly quantify the level of projection effect in these settings and for all cases these values are sufficiently small to continue our analysis.

Next, we look at the impact of the approximations on biases in cosmological constraints. For Limber, it introduces a bias of 0.73σ to Roman-Real and a bias of 0.75σ to Roman-Harmonic, while the number for Rubin LSST-Y10 is larger, at 1.79σ to Rubin-Real and 1.97σ to Rubin-Harmonic, since the constraining power of Rubin LSST-Y10 is greater than that of Roman HLIS-Y5. We note that Limber mainly affects data vectors at large scales, where the scale cut has little impact, as shown in Figures 4 and 5. The errors induced by the orange line (Limber) are completely outside the scale cut in harmonic space, while a small part of it is removed in real space. This also explains why the bias in harmonic space is larger than that in real space.

For noRSD, it introduces a bias of 0.70σ to Roman-Real, 0.75σ to Roman-Harmonic, 2.05σ to Rubin-Real and 2.31σ to Rubin-Harmonic. The bias introduced by noRSD is equivalent and even stronger than that of Limber. This is revealed by the red dotted line of galaxy clustering in Figures 4 and 5 which has equivalent and even greater amplitude compared to yellow line (Limber). The main impact of noRSD lies in the

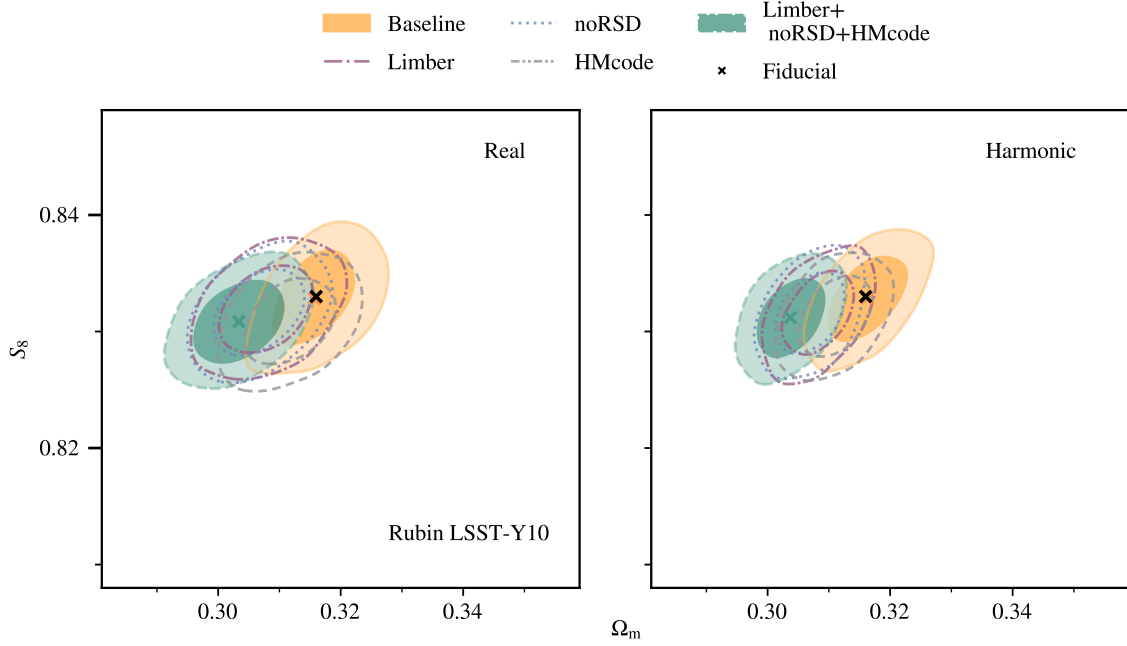


FIG. 7. This plot presents $S_8 - \Omega_m$ constraints of five scenarios as described in Figure 6 for survey Rubin LSST-Y10. The inner and outer contours enclose the area representing 95% and 68% confidence level. The yellow and green cross represents the posterior means of scenario Baseline and the most biased scenario Limber-noRSD-HMCode. The black cross represents the input fiducial value.

large scales, same as the Limber case, where the scale cuts have little influence. The bias is greater for Rubin LSST-Y10 than that for Roman HLIS-Y5, for the same reasons discussed in the Limber case.

For HMCode, it introduces a bias of 0.61σ to Roman-Real, 0.56σ to Roman-Harmonic, 0.98σ to Rubin-Real and 1.24σ to Rubin-Harmonic. These numbers are smaller than those of Limber and noRSD. This is related to the fact that the difference in the data vectors from different nonlinear matter power spectrum prescriptions occurs on small scale, which in our case is removed by scale cuts. This can be seen by the green line in Figures 4 and 5 that scales where it contributes most are covered by our scale cut. Its bias for Rubin LSST-Y10 is also found to be larger than that of Roman HLIS-Y5 due to the constraining power.

For Limber+noRSD+HMCode, the introduced bias significance is the greatest, 0.96σ to Roman-Real, 1.12σ to Roman-Harmonic, 2.76σ to Rubin-Real and 3.12σ to Rubin-Harmonic. This demonstrates how the inferred cosmology could bias if we neglects all of the three modeling effects, though the bias introduced by any single modeling component may already be unacceptable. We note that since HMCode mainly biases data vector at small scale, while noRSD and Limber has similar effect at large scale, their effects do not simply cancel each other.

We also calculate the Figure of Merit (FoM) in the $\Omega_m - S_8$ plane for each survey's Baseline case defined as $\text{FoM} \equiv 1/(\sqrt{\det(\text{Cov}_{\Omega_m-S_8})})$. The FoM thus is a metric of the constraining power that higher value representing more constraint contour while lower for the opposite.

We find a FoM of 29,127 for Roman-Real, 36,784 for

Roman-Harmonic, 93,212 for Rubin-Real, and 111,233 for Rubin-Harmonic. The harmonic space FoM is generally larger than that of the real space in our specific scale cut implementation.

VI. CONCLUSIONS

In this paper, we study three physical modeling approximations in a standard 3×2 pt cosmology analysis: Limber approximation, not including Redshift Space Distortion (RSD), and an imperfect modeling of the nonlinear matter power spectrum. These approximations can bias the data vector and propagate into bias in the inferred cosmological parameters. Some of these approximations have been adopted in Stage-III galaxy surveys, where the statistical power were not high enough for the approximations to impact the final analyses. However, when we move onto Stage-IV surveys such as Roman and Rubin, it is important to revisit these approximations.

We setup experiments for the final 5-year Roman High Latitude Imaging Survey and the final 10-year Rubin Legacy Survey of Space and Time. We use the software package COCOA, which wraps around the cosmology library CosmoLike to carry out our analysis. We cross-check CosmoLike with existing code base CCL and find excellent agreement between the two codes.

We next turn on and off each of the three approximations, and observe the difference in the data vector and the resulting cosmological constraints. We find that using Limber approximation or not including RSD can result in $20 \sim 30 \Delta\chi^2$ for Roman and over $130 \sim 180 \Delta\chi^2$ for Rubin, while the cosmo-

logical bias induced in $\Omega_m - S_8$ plane is about 1σ for Roman and 2σ for Rubin. We also find that using a different nonlinear matter power spectrum prescription, i.e. HMCODE compared EUCLIDEMULATOR2 can result in $3 \sim 7 \Delta\chi^2$ for Roman and $15 \sim 35 \Delta\chi^2$ for Rubin, while the bias is about $0.6 \sim 0.8\sigma$ for Roman and $0.9 \sim 1.3\sigma$ for Rubin. These number can vary under different configurations of the data vector (real or harmonic), scale cuts, and covariance matrix.

Our results have implication for the Stage-IV surveys: 1) It is important to optimize the non-Limber algorithm, so that it could become more numerically stable and efficient. 2) For RSD, our work studied the linear order effect and found a large impact on the data vector and cosmological parameters. This motivates a more realistic RSD model. 3) The choice of model for the nonlinear matter spectrum is important, and there is an interplay of the importance of the model and the chosen scale cuts, where more aggressive scale cuts increases the sensitivity to such choice. We discuss in Appendices B and C how our main results will change when a different lens sample is used and when a different scale cut choice is used.

We imagine several extensions to this work: First, as we have investigated some of the more basic approximations, the next step could be focused on more advanced modeling components such as nonlinear galaxy bias, the intrinsic alignment model, baryonic feedback model etc.. Second, although we have made an attempt to make our survey configurations more realistic by adapting different lens samples and a DES-like scale cut, there are more aspects in our analysis that could be made more realistic based on what we have learned in Stage-III surveys, including the redshift distributions of sources and lenses, and sample selection. Understanding the effect of these simplifications would be important. Finally, our work only used the Roman medium tier for simplicity, while there will also be the wide-tier data, which could increase the statistical power significantly. Investigating on how to combine the two datasets to maximize the power of the Roman survey would be an interesting future direction.

Appendix A: Real-space data vector comparison

We will briefly discuss two details in the real-space code comparison here: the angular bin window function (or, bin-average) and the flat/curve-sky transformation. The former refers to the fact that when we compare with an actual measured data vector, we need to average the correlation function model over a given angular bin $[\theta_{\min}, \theta_{\max}]$ where the data is measured. The latter relates to the exact mathematical transformation from angular power spectrum into the correlation functions.

- *Treatment of averaging the model within angular bin:* In earlier cosmic shear analyses such as DES Y1 [20], the model value for a given angular bin $[\theta_{\min}, \theta_{\max}]$ was approximated by the model value at some effective mean angular value instead of averaging over the angular bin,

$$\bar{\theta} = \frac{2}{3} \frac{\theta_{\max}^3 - \theta_{\min}^3}{\theta_{\max}^2 - \theta_{\min}^2}. \quad (\text{A1})$$

This approximation was made to avoid the computationally expensive numerical average over angular bin and was verified to be sufficient given the DES-Y1 statistical uncertainties. However, as an analytical average approach was employed in DES-Y3, as discussed in Section II A 2, the computational expense is no longer a problem for MCMC. As such, we argue that the effective mean angular value should not be used. To demonstrate this, We setup both surveys using the source and lens sample in Figure 1, calculate the correlation function in COCOA with analytical bin-average while employ the effective mean angular value described by Equation A1 in CCL. The resulting $\Delta\chi^2$ of data vector comparison is dramatic even under our fiducial scale cut which is $\Delta\chi_{\text{fid-cut}}^2(\xi^\pm, \gamma_t, w, 3 \times 2\text{pt})_{\text{Roman}} = (0.399, 1.849, 3.488, 4.998)$ for Roman HLIS-Y5 and $\Delta\chi_{\text{fid-cut}}^2(\xi^\pm, \gamma_t, w, 3 \times 2\text{pt})_{\text{Rubin}} = (0.591, 8.420, 20.653, 26.363)$ for Rubin LSST-Y10.

- *Flat/Curve-sky transformation:* Transforming the angular power spectrum into the correlation function is the last step of data vector modeling in real space. One can assume the observational sky patch is a flat plane to make this transformation mathematically easier and computationally cheaper, because transforming power spectrum into a curved sky involves a summation of power spectrum over multiples weighted by ℓ -dependent Legendre function, which was thought to be computationally expensive, while the former only requires a integration weighted by the Bessel function that can be computed easily. However, this is no longer a barrier for MCMC as long as we cache those Legendre function properly.

We note that some previous work only implement curved-sky transformation for galaxy-galaxy lensing and galaxy clustering, while leaving cosmic shear assuming flat-sky. We find that at least for Roman HLIS-Y5 and Rubin LSST-Y10, this approximation will be insufficient. When comparing the cosmic shear data vector with and without curved-sky transformation under the fiducial settings for Roman and Rubin, we find $\Delta\chi^2(\xi^\pm, \xi_{\text{fid-cut}}^\pm)_{\text{Roman}} = (3.522, 0.374)$ and $\Delta\chi^2(\xi^\pm, \xi_{\text{fid-cut}}^\pm)_{\text{Rubin}} = (6.005, 0.970)$. This demonstrates even with our fiducial scale cut, the $\Delta\chi^2$ introduced by flat-sky against curved-sky is not ignorable for both surveys. The effect will also be larger for more aggressive scale cuts.

Appendix B: Lens samples in existing literature

As discussed in Section III, there has been different assumptions for the Roman lens sample in the literature, while in this paper we choose to use yet another new sample that we deem more realistic. To assess the impact of this choice, we compare the constraining power for our fiducial Roman HLIS-Y5 survey when the different lens samples are used. The four lens samples are:

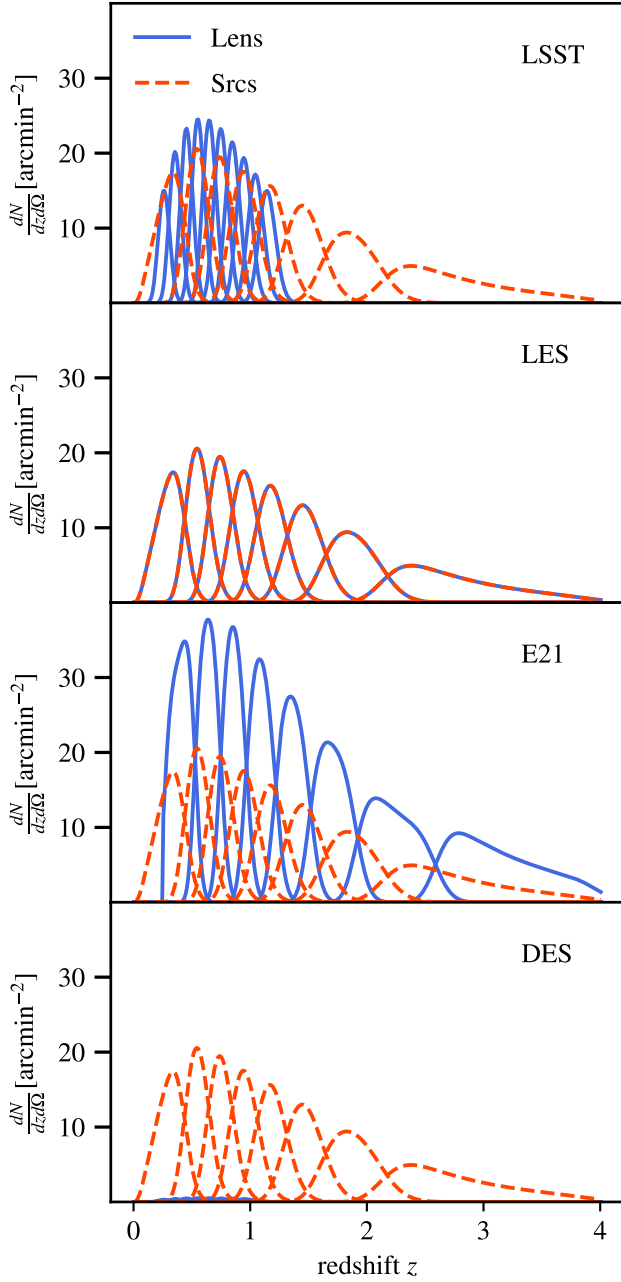


FIG. 8. Lens and source redshift distributions of scenarios discussed in Appendix B.

- **LSST:** This is the fiducial lens sample used in this paper. The sample is defined using the Smail equation with parameterization given by LSST DESC SRD, then divided into 10 tomographic bins uniformly from $z \sim 0.2$ to $z \sim 1.2$. The final effective number density for this lens sample is $n_{\text{eff}} = 30 \text{ arcmin}^{-2}$.
- **LES:** A series of papers [2, 4, 14, 29] have directly adopted the source galaxy sample as the lens galaxy sample, which is said to be beneficial for statistical power by reducing number of nuisance parameters marginalized.

However, the plausibility of this method is still uncertain since it has not been used in real surveys.

- **E21:** In [10], the authors used the Galaxy Survey Exposure Time Calculator (ETC) with CANDELS catalog [16], setting different selection function, to generate the lens and source sample. We replicate that sample following the medium tier definition of Roman HLIS-Y5. We set exposure time $5 \times 10^7 \text{ sec}$ for each pixel and band pass for YJH filters of Roman telescope. Since lens sample does not need shape selection, we set the minimum signal-to-noise ratio (SNR) to 10, maximum ellipticity error σ_e to 0.99, and minimum resolution factor R to 0.01. We fit the selected samples into the Smail equation with a fixed $\beta = 2$, generate the analytical lens sample distribution, then divide the $z > 0.2$ part into 8 tomographic bins with equivalent number density while convolving them with a gaussian kernel $\sigma_z = 0.05$. The resulting lens sample has an effective number density $n_{\text{eff}} = 67 \text{ arcmin}^{-2}$.
- **DES:** We assume a lens sample that has the similar selection function of DES-Y6 while keeping the overall shape of LSST lens redshift distribution. This results in a total number density of $n_{\text{eff}} = 0.633 \text{ arcmin}^{-2}$. We regard this as a conservative scenario in a sense that we can already characterize a lens sample like this and no new methodologies are needed.

Figure 8 illustrates the redshift distribution and number density for each of the lens samples, compared to the fixed source sample. The effective number density for each sample is listed in Table V.

Given these lens samples and the Roman source sample, we thus follow Section IV B to generate the covariance matrix for each scenario, then implement the Fisher analysis following [4] with our fiducial scale cut defined in Section IV A to obtain an upper bound estimation of constraining power. We follow Section V C to calculate the FoM based on $\Omega_m - \sigma_8$ plane for constraints obtained from the Fisher and the results are shown in Table V.

The result shows a general trend that constraining power of $3 \times 2\text{pt}$ and $2 \times 2\text{pt}$ rises with the increase of the lens sample number density. The exception is that the LES case has the largest FoM even though the number density is lower than E21. This is because lens sample and source sample in LES share the same set of photo- z nuisance parameters and less nuisance parameters to marginalize over. We also note that even though number density of lens sample of DES is two order of magnitude lower than those of other scenarios, the FoM of $3 \times 2\text{pt}$ is only 39% less than the E21 case. This demonstrates that constraining power has a diminishing return from the increase of lens sample number density, while the rise of which can lead to lower photo- z accuracy [26].

Table V also makes clear the how the balance between cosmic shear and $2 \times 2\text{pt}$ shifts as a function of the lens sample. As the lens number density increases, constraining power of $2 \times 2\text{pt}$ becomes dominating compared to that of cosmic shear. In the LES scenario, the FoM of $2 \times 2\text{pt}$ is more than 3 times

TABLE V. This table shows FoMs of probes and the number density of lens and source samples for scenarios discussed in Appendix B. FoMs are calculated based on area of $\Omega_m - \sigma_8$ plane with 68% confidence level. The number density is in unit of arcmin^{-2} .

FoMs/ n_{eff}	$3 \times 2\text{pt}$	cosmic shear	$2 \times 2\text{pt}$	lens	source
LSST	28,672	8,123	13,754	30	41
LES	44,751	8,123	28,834	41	41
E21	30,513	8,123	15,884	67	41
DES	18,721	8,123	7,284	0.633	41

than that of cosmic shear. In this scenario, one would imagine that the detailed characterization of the lens sample will become more important.

Appendix C: Impact of scale cut choices

As discussed in Section IV A, our analysis assumes a fiducial scale cut that is similar to what is adopted in DES: a minimum transverse comoving radius ($R_{\text{min}}^{\gamma t}, R_{\text{min}}^w$) = (6Mpc/h, 8Mpc/h) for galaxy-galaxy lensing and galaxy clustering, and a $\Delta\chi^2$ threshold for cosmic shear. This is different from most forecasting papers for both Roman and Rubin. In particular, many Roman and Rubin forecast papers adopt the scale cut choices used in the LSST DESC SRD, where $\ell_{\text{max}} = 3000$ for cosmic shear and $k_{\text{max}} = 0.3h/\text{Mpc}$ for galaxy clustering and galaxy-galaxy lensing. In this case, few points are actually discarded by its scale cut for cosmic shear which emphasizes the assumption that we will be able to model the information of cosmic shear to such small scale.

The $k_{\text{max}} = 0.3h/\text{Mpc}$ is equivalent to a minimum transverse radius $R_{\text{min}} = 21\text{Mpc}/h$ which is more conservative compared to the number used in DES, discarding about 1/2 data points from galaxy-galaxy lensing and galaxy clustering. Here we note our fiducial scale cut as DES-like while the SRD assumed scale cut as SRD-like.

We also introduce two different lens samples, LES and E21, into this comparison of scale cut, against our fiducial lens sample LSST. We thus study six different combinations of lens sample and scale cut: LSST×DES-like, LES×DES-like, E21×DES-like, LSST×SRD-like, LES×SRD-like, and E21×SRD-like. The first scenario is exactly our baseline in this paper, while the last scenario is more close to previous Roman forecast paper lens and source sharing the same galaxy sample.

We then run MCMC assuming the fiducial cosmology in this paper, and the marginal distribution of parameters Ω_m, σ_8, S_8 is shown in Figure 9. It is very clear that our baseline has the least constraining power and using the same galaxy sample for lens and source is beneficial to statistical power. SRD-like scale cut is the dominant factor boosting the constraining power as demonstrated at the rightmost panel of Figure 9. Figure 9 illustrates that neither of these scenarios leads to a noticeable inference bias though their constraining power is distinct, which can result a physical modeling has different priorities using different lens samples and scale cuts. For example, an effect is ignorable in LSST×DES-like can be important in LES×SRD-like due to the latter scenario has much stronger constraining power. This reveals the importance of having the most realistic galaxy sample and scale cut before we determine how an effect is important in the analysis pipeline.

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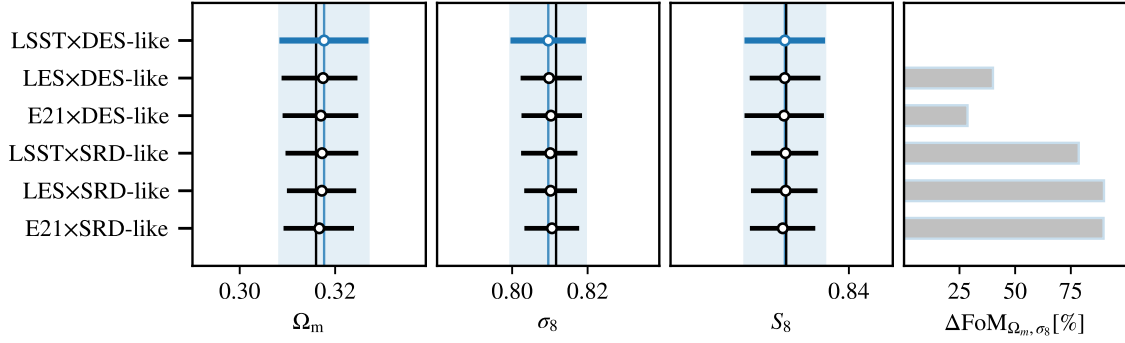


FIG. 9. The left three plots compares the posterior mean of parameters σ_8, Ω_m , and S_8 , under different scenarios which are explained in Appendix C. The vertical black solid line represents the input fiducial, while the blue one is the posterior mean of our baseline, the difference of which shows the level of projection effect. The rightmost plot shows the Figure of Merit of the parameter pair $\sigma_8 - \Omega_m$ under different scenarios.

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